

SHOFIA FAUZIYAH

Course by NordEdu
SQL #10

15 DAYS OF USING
SQL

Mastering Data Retrieval & Management with
PostgreSQL

Challenge

In this challenge, we use again the view v_customer_info that we have previously created:

```
CREATE VIEW v_customer_info
```

```
AS
```

```
SELECT cu.customer_id,
```

```
cu.first_name || ' ' || cu.last_name AS name,
```

```
a.address,
```

```
a.postal_code,
```

```
a.phone,
```

```
city.city,
```

```
country.country
```

```
FROM customer cu
```

```
JOIN address a ON cu.address_id = a.address_id
```

```
JOIN city ON a.city_id = city.city_id
```

```
JOIN country ON city.country_id = country.country_id
```

```
ORDER BY customer_id
```

You need to perform the following tasks:

1) Rename the view to v_customer_information.

2) Rename the customer_id column to c_id.

3) Add also the initial column as the last column to the view by replacing the view.

Challenge

```
1 CREATE VIEW v_customer_info
2 AS
3 SELECT cu.customer_id,
4        cu.first_name || ' ' || cu.last_name AS name,
5        a.address,
6        a.postal_code,
7        a.phone,
8        city.city,
9        country.country
10 FROM customer cu
11 JOIN address a ON cu.address_id = a.address_id
12 JOIN city ON a.city_id = city.city_id
13 JOIN country ON city.country_id = country.country_id
14 ORDER BY customer_id;
```

notes

Dengan membuat tabel view customer information ini mempermudah user untuk mendapatkan detail informasi dari setiap customernya

Challenge

```
16 SELECT *
17 FROM v_customer_info;
```

Data Output Messages Notifications								
Showing rows: 1 to 599 Page No: 1 of 1								
	customer_id integer	name text	address text	postal_code text	phone text	city text	country text	
1	1	MARY SMITH	1913 Hanoi Way	35200	28303384290	Sasebo	Japan	
2	2	PATRICIA JOHNSON	1121 Loja Avenue	17886	8386352866...	San Bernardino	United States	
3	3	LINDA WILLIAMS	692 Joliet Street	83579	4484771904...	Athenai	Greece	
4	4	BARBARA JONES	1566 Inegl Manor	53561	7058140035...	Myingyan	Myanmar	
5	5	ELIZABETH BROWN	53 Idfu Parkway	42399	10655648674	Nantou	Taiwan	
Total rows: 599 Query complete 00:00:00.301 CRLF Ln 10, Col 17								

notes

Mengecek hasil create view dengan menampilkan semua isi tabelnya, dapat dilihat dari output data yang ditampilkan merupakan detail informasi dari setiap customer

```
20  -- 1) Rename the view to v_customer_information.
21  ALTER VIEW v_customer_info
22  RENAME TO v_customer_information;
23
```

Data Output Messages Notifications

ALTER VIEW

Query returned successfully in 145 msec.

Challenge

```
24  SELECT *
25  FROM v_customer_information;
```

Data Output Messages Notifications

Showing rows: 1 to 599 Page No: 1 of 1

	customer_id integer	name text	address text	postal_code text	phone text	city text	country text
1	1	MARY SMITH	1913 Hanoi Way	35200	28303384290	Sasebo	Japan
2	2	PATRICIA JOHNSON	1121 Loja Avenue	17886	8386352866...	San Bernardino	United States
3	3	LINDA WILLIAMS	692 Joliet Street	83579	4484771904...	Athenai	Greece
4	4	BARBARA JONES	1566 Inegl Manor	53561	7058140035...	Myingyan	Myanmar
5	5	ELIZABETH BROWN	53 Idfu Parkway	42399	10655648674	Nantou	Taiwan

Total rows: 599 Query complete 00:00:00.303 CRLF Ln 22, Col 34

notes

Mengubah nama tampilan file customer view, ini dapat memudahkan kita sebagai user jika ingin menggunakan data kembali namun dengan tujuan atau format yang berbeda


```
30 -- 2) Rename the customer_id column to c_id.
31 ALTER VIEW v_customer_information
32 RENAME COLUMN customer_id TO c_id;
```

Data Output Messages Notifications

ALTER VIEW

Query returned successfully in 140 msec.

Challenge

```
37 SELECT *
38 FROM v_customer_information;
```

Data Output Messages Notifications

Showing rows: 1 to 599 Page No: 1 of 1

	c_id integer	name text	address text	postal_code text	phone text	city text	country text
1	1	MARY SMITH	1913 Hanoi Way	35200	28303384290	Sasebo	Japan
2	2	PATRICIA JOHNSON	1121 Loja Avenue	17886	8386352866...	San Bernardino	United States
3	3	LINDA WILLIAMS	692 Joliet Street	83579	4484771904...	Athenai	Greece
4	4	BARBARA JONES	1566 Inegl Manor	53561	7058140035...	Myingyan	Myanmar
5	5	ELIZABETH BROWN	53 Idfu Parkway	42399	10655648674	Nantou	Taiwan

Total rows: 599 Query complete 00:00:00.513 CRLF Ln 33, Col 1

notes

Mengubah vie w hanya pada salah satu kolom saja, dapat memudahkan user jika memerlukan penyesuaian data dengan tujuan penggunaan data yang berbeda juga

Challenge

```
40 -- 3) Add also the initial column as the last column to the view by replacing the view.  
41 ALTER TABLE v_customer_information  
42 ADD COLUMN initial VARCHAR(10);
```

Data Output Messages Notifications

ERROR: ALTER action ADD COLUMN cannot be performed on relation "v_customer_information"
This operation is not supported for views.

SQL state: 42809

Detail: This operation is not supported for views.

notes

Pada pervobaan ini, mencoba langsung menambahkan kolom initial pada tabel view customer, namun terdapat error message dimana "operation is not suppoeted for views. Kaerena pada tabel view itu bukan tabel asli, melainkan hasil dari query yang hanya bisa menampilkan format data tanpa merubah value

Challenge

```
Query  Query History
44  -- drop view existing
45  DROP VIEW IF EXISTS v_customer_info;
46
47  -- recreate table dan tambahkan kolom initial
48  CREATE VIEW v_customer_info AS
49  SELECT |
50      cu.customer_id,
51      cu.first_name || ' ' || cu.last_name AS name,
52      a.address,
53      a.postal_code,
54      a.phone,
55      city.city,
56      country.country,
57      LEFT(cu.first_name, 1) AS initial -- kolom tambahan
58  FROM customer cu
59  JOIN address a ON cu.address_id = a.address_id
60  JOIN city ON a.city_id = city.city_id
61  JOIN country ON city.country_id = country.country_id
62  ORDER BY customer_id;
63

Data Output  Messages  Notifications
CREATE VIEW
Query returned successfully in 161 msec.
```

notes

Jadi jika ingin menambah kolom initial kita perlu drop terlebih dahulu, Karena berbeda dengan tabel, view tidak bisa diubah strukturnya dengan perintah ALTER TABLE lalu ADD COLUMN. agar view punya struktur sesuai kebutuhan.

Challenge

notes

Mengecek penambahan kolom initial yang diambil dari kolom name atau lenih tepatnya first_name yang nantinya dapat memudahkan user jika ingin mengelompokan/grouping berdasarkan initial user, dan pengolahan data lainnya

64
65

SELECT initial

FROM v_customer_info;

Data Output

Messages

Notifications

≡+

📄

▼

📋

▼

🗑

🗄

⬇

📈

SQL

	initial text
1	M
2	P
3	L
4	B
5	E
6	J

Total rows: 599

Query complete 00:00:00.254

64
65

SELECT *

FROM v_customer_info;

Data Output

Messages

Notifications

≡+

📄

▼

📋

▼

🗑

🗄

⬇

📈

SQL

Showing rows: 1 to 599

Page No: 1 of 1

	customer_id integer	name text	address text	postal_code text	phone text	city text	country text
1	1	MARY SMITH	1913 Hanoi Way	35200	28303384290	Sasebo	Japan
2	2	PATRICIA JOHNSON	1121 Loja Avenue	17886	8386352866...	San Bernardino	United States
3	3	LINDA WILLIAMS	692 Joliet Street	83579	4484771904...	Athenai	Greece
4	4	BARBARA JONES	1566 Inegl Manor	53561	7058140035...	Myingyan	Myanmar
5	5	ELIZABETH BROWN	53 Idfu Parkway	42399	10655648674	Nantou	Taiwan

Total rows: 599

Query complete 00:00:00.347

CRLF

Ln 59, Col 47

THANKYOU